



The role of executive functions and fluid intelligence in the prediction of conditional strategy knowledge of self-regulated learning

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ARTICLE INFO

Keywords:

Self-regulated learning
Executive functions
Intelligence
Preschool children
Elementary school children

ABSTRACT

Self-regulated learning (SRL) is an important predictor for academic achievement already at preschool and elementary school age. Conditional strategy knowledge of SRL is considered as a central prerequisite for the use of SRL strategies. Individual differences in SRL in same-aged children may be attributed to differences in cognitive abilities. The present study investigated the contribution of executive functions (EF) and fluid intelligence to conditional strategy knowledge of SRL in a sample of 5- to 6-year-old ($n = 259$) and 8- to 12-year-old children ($n = 211$). Structural equation modeling (SEM) revealed that EF performance significantly predicted conditional strategy knowledge of SRL in both age groups whereas fluid intelligence was no significant predictor, suggesting that executive functions, but not fluid intelligence, are associated with conditional strategy knowledge of SRL in childhood. These results complement findings on the relations of executive functions, intelligence, and strategy usage and may inform interventions aimed at fostering children's learning behavior.

1. Introduction

In information society with its demand to continuously acquire new knowledge and skills, self-regulated learning (SRL) is considered an important competence (Zeidner & Stoecker, 2019). SRL includes processes whereby learners regulate and monitor their cognition, motivation, and behavior towards the attainment of learning goals (Pintrich, 2000; Zimmerman & Schunk, 2011). It is a significant predictor of academic achievement in formal learning environments including elementary and secondary school (Dent & Koenka, 2016). Even preschool-aged children can regulate their learning and benefit from it with regard to their academic competence (Bronson, 2000; Grüneisen et al., 2023) stressing the relevance of SRL for educational research and practice.

Theoretical models propose a range of skills that are relevant for successful SRL (e.g., Boekaerts, 1999). Among those, strategy knowledge of SRL is considered a central competence (Dresel et al., 2015). Moreover, individual differences in cognitive abilities might affect SRL (Hoyle & Dent, 2018). The present study thus investigated the interplay of conditional strategy knowledge of SRL, executive

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functions (EF), and fluid intelligence in preschool and elementary school children aged 5–6 years and 8–12 years, respectively.

1.1. Self-regulated learning

There are numerous theoretical conceptualizations of SRL which converge on the notion that the construct is characterized by intellectual capabilities and affective-motivational states (see [Panadero, 2017](#); [Tinajero et al., 2024](#) for reviews). However, they differ in the emphasis they place on specific components as well as cyclic phases of SRL. [Wirth and Leutner \(2008\)](#) thus formulated the categorization into component and process models of SRL (see also [Thillmann, 2007](#)). Component models describe different competencies or learner attributes that enable successful SRL. One prominent example is the three-layer model proposed by [Boekaerts \(1999\)](#). In this model, SRL is regarded as a complex interaction of cognitive, metacognitive, and motivational abilities displayed as three concentric layers. The core layer (i.e., cognitive layer) entails the regulation of information processing by selecting from a repertoire of cognitive learning strategies such as elaboration and organization of learning material (e.g., [Pintrich et al., 1993](#)). The middle layer (i.e., metacognitive layer) includes metacognitive knowledge and strategies such as planning, monitoring, and evaluation of the learning process ([Boekaerts, 1999](#)). The outer layer (i.e., motivational layer) refers to the regulation of the self and includes motivational and volitional processes elicited for goal attainment.

In contrast, process models conceptualize SRL as a sequence of different phases and account for its cyclical nature ([Klug et al., 2011](#); [Wirth & Leutner, 2008](#)). The social-cognitive model of self-regulation ([Zimmerman, 2000](#); [Zimmerman & Moylan, 2009](#)) and adapted versions (e.g., [Schmitz & Wiese, 2006](#)) describe SRL in terms of three successive phases including a planning phase preceding learning, an action phase covering learning, and a reflection phase following when learning has ended. Conclusions drawn from the reflection phase can inform the planning phase of subsequent learning actions. Additionally, cognitive (e.g., task-related strategies), metacognitive (e.g., self-monitoring), and motivational (e.g., volitional strategies) processes are required within the three phases, thus reflecting the components described in [Boekaerts' \(1999\)](#) and other models of SRL (e.g., [Pintrich, 2000](#)).

1.2. Development of self-regulated learning

With regard to the age-related maturation of SRL, [Hoyle and Dent \(2018\)](#) argue that the construct encompasses a multitude of abilities that together enable the development of SRL. Following this perspective, we describe the emergence and maturation of knowledge and skills subsumed under the SRL components of cognition, metacognition, and motivation ([Boekaerts, 1999](#)) across the preschool and elementary school years.

The cognitive component of SRL entails the selection and application of a range of learning strategies, such as rehearsal, elaboration, and organizational strategies (e.g., [Pintrich et al., 1993](#)). During the preschool years, children rely on simple rote-type strategies such as attending to items they would like to remember ([Bjorklund et al., 2009](#); [Schneider, 2014](#)). Rehearsal and organizational strategies are still inefficient and rarely applied spontaneously. At elementary school age, children use rehearsal and categorization more frequently and efficiently, and also combine multiple strategies (P. A. [Alexander et al., 1998](#); [Kron-Sperl et al., 2008](#)). More complex strategies, such as elaboration and deep-processing techniques, continue to develop during adolescence and are acquired mainly through formal instruction across the secondary school years ([Bjorklund et al., 2009](#)).

The metacognitive component of SRL encompasses metacognitive knowledge and strategies such as goal-setting, planning, self-monitoring, self-control, and self-evaluation ([Dent & Koenka, 2016](#)). Preschool-aged children are able to adopt learning goals, yet these appear rather fragile when situational demands change ([Leclercq et al., 2023](#)). Moreover, they possess rudimentary metacognitive knowledge on learning tasks and strategies ([Schneider, 2008](#); [Schneider & Lockl, 2006](#)). Basic metacognitive skills allow for monitoring the progress on age-appropriate tasks and evaluation of the outcome ([Bryce & Whitebread, 2012](#)). However, monitoring of performance is overly optimistic ([Lipko et al., 2009](#); [Lipowski et al., 2013](#)), and transfer to adequate control behaviors is only beginning to develop at this age ([Destan et al., 2014](#); [Schneider, 2008](#)). Across the elementary school years, metacognitive knowledge on tasks and strategies increases steadily, which supports children's goal-setting and action planning ([Schneider, 2008](#); [Schneider & Lockl, 2006](#)). Metacognitive monitoring judgments of children's performance become relatively accurate at 8 years of age ([Roebers, 2014](#)). The control of learning behavior based on these monitoring judgments increases towards the end of the elementary school years and continues to develop across adolescence ([Roebers, 2014](#); [Schneider, 2008](#)).

The component of motivation entails motivational beliefs and attitudes, as well as strategies aimed at regulating motivation ([Wolters, 2003](#)). Preschool-aged children have high self-efficacy beliefs ([Wigfield et al., 2011](#)) and persist at challenging tasks ([Mokrova et al., 2013](#)). Yet, their knowledge of strategies to regulate motivation is restricted to concrete strategies such as eliminating distractors in the environment while working on a task ([Cooper & Henderlong Corpus, 2009](#)). During the elementary school years, a decline in learning motivation can be observed ([B. Spinath & F. M. Spinath, 2005](#)). As children receive normative feedback and engage in social comparison, their intrinsic motivation decreases, and their self-efficacy beliefs become more congruent with actual performance ([Schunk & Pajares, 2002](#); [B. Spinath & Steinmayr, 2008](#)). At the same time, children gain knowledge on mental strategies to sustain motivation, such as self-talk and interest enhancement, which enables them to actively influence their motivational state when facing difficulty in task completion ([Cooper & Henderlong Corpus, 2009](#)).

In summary, children at preschool age become increasingly able to approach learning tasks in a goal-directed way. Further improvements in abilities related to SRL during middle childhood enable children at elementary school age to regulate their learning more effectively. Yet, there are individual differences in SRL competencies among same-aged children which may be attributed to both individual as well as contextual factors ([Perry et al., 2018](#); [Wigfield et al., 2011](#)).

1.3. Individual differences and self-regulated learning

Correspondingly, some theorists have differentiated the role of person- and context variables in theoretical accounts of SRL. Winne and Hadwin (1998) define conditions which affect other components of the learning cycle. Whereas external conditions refer to environmental features of a learning event such as time constraints and available resources, internal conditions include individual differences in motivational orientations, content knowledge, metacognitive knowledge of task strategies, and cognitive capacity (Winne, 2011; Winne & Hadwin, 1998). Similarly, an explicit distinction of person variables which guide self-regulation at the task-level was proposed in the metacognitive and affective model of self-regulated learning (MASRL; Efklides, 2011; Efklides & Schwartz, 2024). It proposes a range of relatively stable person characteristics, including cognitive abilities, metacognitive knowledge and skills, motivation, and affect. These variables interact with each other at the Person level, for example, cognitive abilities are related to a person's metacognitive knowledge of the self which in turn affects motivational orientations including expectancies for success or failure (Efklides, 2011). The interaction of Person-level variables affects how individuals approach learning situations or specific tasks. At the Task x Person level, the MASRL model proposes cognitive processes and metacognitive experiences, such as feelings of ease or difficulty, that guide task-processing and inform the Person level in a bottom-up process (Efklides, 2011). While a comprehensive description of these models is beyond the scope of this article, the emphasis they place on person variables with regard to SRL can inform research on individual differences that affect the development of SRL (Efklides & Schwartz, 2024). The current study thus focuses on knowledge about cognitive, metacognitive, and motivational strategies as well as cognitive abilities such as executive functions and fluid intelligence, which have been proposed as candidate variables associated with the development of SRL (Dresel et al., 2015; Hoyle & Dent, 2018).

1.4. Strategy knowledge of self-regulated learning

The application of cognitive, metacognitive, and motivational strategies involved in SRL can be conceptualized as enacted knowledge about such strategies (Dörrenbächer-Ulrich et al., 2024). Dresel et al. (2015) consider strategy knowledge of SRL a central competence with regard to the execution of cognitive, metacognitive, and motivational strategies involved in SRL. An important distinction concerns the type of strategy knowledge, with declarative strategy knowledge referring to knowledge on what a strategy is and procedural strategy knowledge referring to knowledge on how to apply a specific strategy (Hasselhorn & Artelt, 2018). However, the mere knowledge about cognitive, metacognitive, and motivational strategies and how these are executed does not seem sufficient for their successful application (Dresel et al., 2015). Therefore, the term conditional strategy knowledge was introduced, referring to knowledge on the suitability of a strategy given specific situational demands (Dörrenbächer-Ulrich et al., 2024; Paris et al., 1983). Empirically, conditional strategy knowledge of SRL has been identified as a predictor of the application of (meta-)cognitive strategies in adolescents (Karlen, 2015). In children, the assessment of conditional strategy knowledge of SRL as a competence seems especially valuable since cognitive, metacognitive, and motivational strategies might not yet be fully integrated into overt behavior (Grüneisen et al., 2024).

1.5. Executive functions and fluid intelligence

Executive functions refer to higher-order mental processes that enable individuals to regulate cognition and action in goal-directed behavior (Friedman & Miyake, 2017; Karbach & Unger, 2014). They have been of major interest in psychological research due to their significant implications for developmental outcomes including academic achievement (Best et al., 2011; Blair & Razza, 2007) and social functioning (Riggs et al., 2006). Current conceptualizations of executive functions propose three interrelated, yet separable components (Diamond, 2013, 2020; Miyake et al., 2000). Working memory allows for the short-term storage and manipulation of information in mind. Inhibition refers to the control of attention in light of distracting stimuli in the environment (i.e., interference control) as well as the suppression of a prepotent verbal or motor response in favor of a more appropriate reaction. The third component of EF, cognitive flexibility, is the ability to flexibly switch attention between rules, tasks, or mental sets. The development of these abilities follows differential trajectories. Working memory and inhibition emerge in infancy and improve across childhood and adolescence (Best & Miller, 2010; Diamond, 2013, 2020; Garon et al., 2008), whereas cognitive flexibility follows a protracted developmental course from preschool age to adolescence drawing on both working memory and inhibition. This developmental pattern is associated with the relatively slow maturation of the prefrontal cortex, which represents the neurophysiological substrate of executive functions in childhood (Moriguchi, 2017).

Executive functions are closely related to fluid intelligence (Blair, 2006). By definition, fluid intelligence encompasses abilities of reasoning, problem-solving, and planning to adapt to changing conditions (Duggan & Garcia-Barrera, 2015). Diamond (2013) conceptualized these abilities as higher-level executive functions that are supported by the core components of working memory, inhibition, and cognitive flexibility. Similar to executive functions, fluid intelligence is an important predictor for developmental outcome variables, such as school achievement (Kriegbaum et al., 2018; Roth et al., 2015). In preschool- and elementary school-aged children, strong associations between executive functions and fluid intelligence were reported by several studies (Brydges et al., 2012; Ger & Roebers, 2023; Grobe et al., 2024; Masten et al., 2012). In late childhood, differential relations were found with working memory showing stronger associations with fluid intelligence than inhibition and cognitive flexibility (Duan et al., 2010). In summary, theoretical accounts and empirical findings on the relation between executive functions and fluid intelligence suggest that the two constructs are closely related, yet separable in the late preschool and elementary school years. A comprehensive approach to understanding the interplay of SRL competencies with cognitive processes should thus include both executive functions and fluid

intelligence.

1.6. Executive functions and self-regulated learning

From a theoretical perspective, executive functions are required in the execution of cognitive, metacognitive, and motivational strategies (Pintrich & Zusho, 2002). Accordingly, a number of studies have analyzed the relation between executive functions and enacted SRL in children suggesting positive links between the constructs (e.g., Davis et al., 2021; Ferreira et al., 2022; Grüneisen et al., 2023; Rutherford et al., 2018). Using performance-based tasks of inhibition, cognitive flexibility, and working memory, Davis et al. (2021) examined the concurrent and longitudinal associations between executive functions and teacher-reported SRL by means of structural equation modeling (SEM). In children transitioning from kindergarten to grade 1, they obtained a model in which executive functions and SRL were strongly related ($\beta = .77$). In addition, executive functions in kindergarten significantly predicted SRL in grade 1 ($\beta = .58$). Reversely, kindergarten SRL did not contribute to the prediction of executive functions in grade 1. These results support the notion that executive functions and SRL are strongly related constructs in young children and that executive functions may be a prerequisite for the development of SRL and not vice versa. Results by Rutherford et al. (2018) corroborate these findings. In their sample of elementary school children, performance on a combined inhibition/cognitive flexibility measure in grade 2 was a significant predictor of teacher-reported SRL in grade 3. In addition, SRL partially mediated the relation between executive functions and math abilities. Surprisingly, working memory was not significantly associated with SRL which may be explained by the restricted range and low reliability of the respective measure (Rutherford et al., 2018). In contrast, a latent factor of working memory was associated with latent variables of SRL ($\beta = .40$) and problem-solving in math ($\beta = .49$) assessed with a game-based problem-solving task among third- and fourth-grade elementary school children (Ferreira et al., 2022). Again, there was a significant indirect effect of working memory via SRL on problem-solving in math stressing the importance of both constructs for academic achievement in childhood. While these results suggest an overlap between executive functions and enacted SRL in children, the evidence regarding potential links between executive functions and strategy knowledge of SRL is scarce. In fact, we know of only one study that addressed the role of executive functions in the development of strategy knowledge. Using performance-based measures of executive functions and a scenario-based interview assessing knowledge of memory strategies, Geurten et al. (2016) reported a mediating role of working memory and verbal fluency in the relation between age and internal strategy knowledge in children aged 4–11 years. While this study offers some insight into how executive functions are involved in strategy knowledge, the results are restricted to knowledge of memory strategies (e.g., rehearsal) and are hardly generalizable to children's conditional knowledge of cognitive, metacognitive, and motivational strategies. Moreover, no previous studies have extended their analyses by including other higher-order cognitive variables potentially contributing to the execution or knowledge of cognitive, metacognitive, and motivational strategies, such as fluid intelligence.

1.7. Fluid intelligence and self-regulated learning

Problem-solving, reasoning, and planning abilities associated with the construct of fluid intelligence are important for the development of knowledge and skills in a wide range of academic subjects (Thorsen et al., 2014). Thus, fluid intelligence may also play a role in knowledge and skills related to SRL. However, the empirical evidence on the relations between intelligence and SRL competences during childhood is scarce and rather inconsistent. When enacted SRL was under study, some studies found more frequent use of cognitive, metacognitive, and motivational strategies in highly intelligent youths compared with average-intelligent peers (Bouffard-Bouchard et al., 1993; Zimmerman & Martinez-Pons, 1990), whereas other studies do not report evidence for such an advantage (Dresel & Haugwitz, 2005; Sontag et al., 2012). In fact, when trained to use an organizational strategy, both elementary school children with high and low intelligence increased their use of this learning strategy in subsequent memory tasks (Bjorklund et al., 1994). Yet, highly intelligent children who did not receive any training also acquired an organizational strategy by mere exposure to study material, suggesting that these children spontaneously applied memory strategies. This was not evident for less intelligent children, indicating that intelligence is involved in strategy development. However, its association with strategy use may be buffered by instructional practices. In contrast, Dresel and Haugwitz (2005) found significant negative effects of intelligence on self-reported use of (meta-)cognitive strategies in sixth-grade students during math class. That means, highly intelligent students reported low levels of (meta-)cognitive strategy use. In contrast, motivational variables, including control beliefs and task value, positively predicted (meta-)cognitive strategy use. With regard to the association between intelligence and SRL in younger children, Sontag et al. (2012) obtained similar results. In a large sample of fourth-grade elementary school children, intelligence did not predict the preference for SRL significantly at the beginning of fourth grade. It was also not predictive of the change in the preference for SRL across the school year. Instead, learning goal orientation significantly predicted an increase in the implementation of learning strategies, stressing the importance of motivation for SRL.

The inconsistent findings on the relations between intelligence and cognitive, metacognitive, and motivational strategy use may be explained in light of methodological differences. For example, the conceptualization and operationalization of both intelligence and SRL varied across studies (Sontag et al., 2012; Zeidner & Stoeger, 2019). Several studies included highly specific samples of gifted youths (e.g., Ablard & Lipschultz, 1998) and others selectively focused cognitive strategies such as memory strategies (Bjorklund et al., 1994) limiting the generalizability of results. Moreover, there is a lack of evidence on the relation between intelligence and SRL in preschool- and elementary school-aged children due to a stark underrepresentation of younger aged samples (Stoeger & Sontag, 2012).

Research on associations between intelligence and metacognition (see J. M. Alexander et al., 1995; Ohtani & Hisasaka, 2018 for reviews) may inform the present study as metacognition is considered integral in models of SRL (e.g., Boekaerts, 1999). Moreover, the construct includes metacognitive knowledge (Efklides, 2008), a construct similar to conditional strategy knowledge of SRL. We thus

report on select studies while acknowledging that the conceptualization of metacognitive knowledge as knowledge about cognitive and metacognitive strategies is not fully congruent with the broader conceptualization of conditional strategy knowledge of SRL. The latter explicitly includes knowledge about the suitability of cognitive, metacognitive, and motivational strategies given situational demands (Dörrenbächer-Ulrich et al., 2024) and is theoretically rooted in models of SRL (e.g., Zimmerman, 2000).

Empirical evidence suggests significant positive associations between intelligence and metacognitive skills such as planning and monitoring in adolescents and elementary school-aged children (Desoete, 2008; Snyder et al., 2011; van der Stel & Veenman, 2008, 2014). A meta-analysis synthesizing multiple studies on the relations of intelligence, metacognition, and academic achievement corroborates these findings (Ohtani & Hisasaka, 2018). The authors reported moderate to strong associations between intelligence and metacognition (i.e., metacognitive knowledge and skills) in different age groups including preschool- and elementary school-aged children. Studies focusing on the relation between intelligence and metacognitive knowledge suggest positive links in children of different ages (J. M. Alexander et al., 1995). In a study by Schwanenflugel et al. (1997), highly intelligent children from kindergarten and first grade were more knowledgeable of memory and attention processes than average-intelligent children of the same age. Specifically, intelligence was associated with a greater understanding of why certain strategies were helpful for memorization and how situational influences would impact attentional processes. However, the difference in metacognitive knowledge between highly intelligent and average-intelligent children was rather small. In a different study, children aged 6–9 years with higher intelligence had better knowledge of the usefulness of a category sorting strategy on a sort-recall task than same-aged children of average intelligence (J.M. Alexander & Schwanenflugel, 1994). While the association between intelligence and metacognitive knowledge was moderate, intelligence did not play a central role for the use of memory strategies.

1.8. The present study

Different lines of research have suggested executive functions and intelligence as predictors of SRL (Hoyle & Dent, 2018; Stoeger & Sontag, 2012). However, evidence on the relations of these variables in children is scarce. Most studies have analyzed the relations between executive functions, intelligence and enacted SRL (e.g., Davis et al., 2021; Dresel & Haugwitz, 2005). Conditional strategy knowledge of SRL is considered an important SRL competence, but has thus far not been analyzed in relation to executive functions and fluid intelligence (Dresel et al., 2015). Taking this aspect of SRL into consideration seems especially informative in early and middle childhood since cognitive, metacognitive, and motivational strategies may not have been fully integrated into overt behavior at this point (Grüneisen et al., 2024). Therefore, we sought to investigate the interplay of these constructs at preschool and elementary school age.

Specifically, the present study sought to examine the shared and unique contributions of executive functions and fluid intelligence to conditional strategy knowledge of SRL in a sample of 5- to 6-year-old and 8- to 12-year-old children. We expected that (1) the predictors of executive functions and fluid intelligence would be strongly associated in both age groups and that (2) executive functions would be a significant predictor of conditional strategy knowledge of SRL in both 5- to 6-year-old and 8- to 12-year-old children. (3) Given the scarce and inconsistent findings on the relation between intelligence and SRL competencies in previous research, the role of fluid intelligence as a predictor of conditional strategy knowledge of SRL in 5- to 6-year-old and 8- to 12-year-old children was explored.

Table 1
Demographic variables for both 5- to 6-year-old and 8- to 12-year-old children

Demographic variables	5- to 6-year-old children		–	8- to 12-year-old children	
	<i>n</i>	%		<i>n</i>	%
Region of birth ^a					
Western Europe	253	97.68		196	92.89
Northern Europe	1	0.39			
Eastern Europe	1	0.39		5	2.37
Southern Europe	1	0.39		1	0.47
Western Asia	1	0.39		5	2.37
Southern Asia				1	0.47
Western Africa	1	0.39		1	0.47
Eastern Africa	1	0.39		1	0.47
South America				1	0.47
Main language					
German	248	95.75		198	93.84
Other	11	4.25		13	6.16
Parental education ^b					
Tertiary education	132/147	55.70/64.47		114/101	57.29/54.89
Post-secondary non-tertiary education	86/67	36.29/29.39		73/68	36.68/37.00
No post-secondary education	19/14	8.02/6.14		12/15	6.03/8.15

Note. Parental education is displayed for both parental figures (parent 1/parent 2).

^a The classification of geographical regions is according to UNCTAD (2025).

^b The classification of educational qualifications is according to Eurostat, OECD & UNESCO Institute for Statistics (2015).

2. Methods

2.1. Participants

Since data collection was conducted in two waves (i.e., the school years of 2022/2023 and 2023/2024), analyses on data of a subsample from the first wave (2022/2023) have already been published (Grobe et al., 2024). The present study included $n = 175$ preschool children and $n = 135$ fourth-grade elementary school children from the first wave of data collection and $n = 130$ preschool children and $n = 147$ fourth-grade elementary school children from the second wave of data collection. Thus, the final sample included $n = 305$ preschool children and $n = 282$ fourth-grade elementary school children. Participants were recruited from 29 daycare facilities and 39 elementary schools. Thereof, 95.59 % were public and 4.41 % were private. The majority of daycare facilities and schools were located in semi-rural and rural regions (63.23 %), the remainder was located in cities (36.76 %). Parents provided sociodemographic information, including the child's country of birth, main language spoken at home, and a proxy for socio-economic status (SES) by indicating their highest educational and vocational qualification (see Table 1). SES was higher in the current sample than in the average population, with a higher percentage of parents with tertiary education and a smaller percentage without post-secondary education. A numerical code assigned to each participant ensured complete anonymization of the data. Children were excluded from all analyses due to diagnoses of developmental or neurological disorders that could affect cognition and learning behaviors ($n = 41$), color vision deficiency ($n = 5$), a language barrier ($n = 16$), or missing of testing sessions in a 4-week testing period ($n = 26$). All other children excluded from analyses had no information on the demographic variables of interest ($n = 29$). The final sample consisted of $N = 470$ children including $n = 259$ preschool children (mean age_{years;months} = 5;9, SD_{months} = 5, age range_{years;months} = 5;0 – 6;10, 53.54 % female) and $n = 211$ elementary school children (mean age_{years;months} = 9;10, SD_{months} = 5, age range_{years;months} = 8;10 – 12;1, 49.51 % female). The relatively large age range in the 8- to 12-year-old children recruited from grade 4 in elementary school is common in the German school system (Lange-Küttner, 2025).

2.2. Procedure

Prior to testing, parents provided written informed consent for their child's study participation. Children assented verbally to participate in the study on the first day of assessments. Testing was conducted in a separate room at children's daycare facility or elementary school by trained student assistants following a standardized test protocol. The 5- to 6-year-old children were tested in four sessions à 25–45 min, and the 8- to 12-year-old children were tested in two sessions à 70–90 min. The order of executive function task domains was counterbalanced across testing sessions to avoid sequence effects. The assessment also included tasks not analyzed in the present study. In both age groups, testing sessions were distributed across four weeks at the most.

2.3. Measures

Intelligence and conditional strategy knowledge of SRL were assessed with one measure, respectively, which were completed in an online survey platform. Task instructions and text-based scenarios of the conditional strategy knowledge measure were read out to participants by a research assistant. Measures of executive functions included two instruments assessing working memory, inhibition, and cognitive flexibility, respectively. Application of two measures for each component allowed for complex modeling approaches, including supplementary analyses of differential associations between executive function components and conditional strategy knowledge of SRL at the latent level. Executive function tasks were implemented in an application created with Unity 2020.3.17f1 and included automated audio instructions. Additional verbal instructions were given by a research assistant in predefined order. All tasks were completed on Lenovo Tab M10 3. Gen tablets with 10.1" and 1920/1200 px displays. In order to increase motivation and task engagement, all tasks were framed by an engaging cover story.

2.3.1. SRL: Strategy-knowledge test for SRL (Dörrenbächer-Ulrich et al., 2024)

Conditional strategy knowledge of SRL was assessed with an age-adapted version of the SKT-SRL (Dörrenbächer-Ulrich et al., 2024). The theoretical foundation of this test is an integrative account of component (Boekaerts, 1999) and process models (Zimmerman, 2000; Zimmerman & Moylan, 2009) of SRL. Thus, it assesses conditional strategy knowledge of cognitive, meta-cognitive, and motivational strategies across the planning, action, and reflection phase of a prototypical learning process. On this test, children were presented with scenarios covering typical learning problems in the respective age groups of preschool and elementary school children. For each scenario, children were to rate the usefulness of strategies on a 4-point Likert scale from 1 = *not useful at all* to 4 = *very useful*. In the original version of the task, the scenarios were assigned to the SRL components and phases by experts in the field of self-regulated learning to evaluate their content validity. Then, the same experts rated six associated strategies with regard to their usefulness for each scenario resulting in three useful and three less useful strategies per scenario. The scenarios and strategies were then adapted for preschool- and elementary school-aged children. In the adaptation process, the following aspects were considered. First, the scenario content was selected to be close to children's everyday lives. This ensured that the words used were part of children's passive vocabulary and they could easily imagine the scenarios. Second, the sentence structure was kept simple, for example by using active voice wordings. Third, to reduce cognitive load, each strategy was rated instantly after being read out, and only four strategies were presented in the 5- to 6-year-old children. Fourth, the scenario content and strategies were discussed with experts in the field of developmental and educational psychology to ensure their suitability for the target age groups.

The test started with the presentation of one introductory scenario along with three strategies to be rated to familiarize children

with the task. Then, seven scenarios were presented covering cognitive, metacognitive, and motivational learning problems within the planning, action, and reflection phases, respectively. For each scenario, the 5- to 6-year-old children were to rate two useful and two less useful strategies, whereas the 8- to 12-year-old children were to rate three useful and three less useful strategies. Children rated the usefulness of the strategies by tapping the respective number of stars presented next to each strategy representing the four levels of the Likert scale.

Pairwise comparisons of the ratings of each useful with each less useful strategy resulted in four difference scores in the 5- to 6-year-old and nine difference scores in the 8- to 12-year-old children for each scenario (see Händel et al., 2013; Maag Merki et al., 2013 for a similar approach). These were coded 0 when a useful strategy was rated lower than or equal to a less useful strategy. When a useful strategy was rated 1 point higher than its less useful counterpart, difference scores were coded 1. When the difference in ratings of a useful vs. less useful strategy equaled 2 or 3 points, the difference score was coded 2. Summed scores were calculated from these values for each scenario and served as dependent variables representing conditional strategy knowledge of a specific component (motivation, metacognition, cognition) within a specific phase (planning, action, reflection) of SRL. Cronbach's α of the SKT-SRL scales ranged between .66 and .79 in the 5- to 6-year-old and .70 and .83 in the 8- to 12-year-old children indicating acceptable to good reliability of most scales in both age groups. For an example scenario and associated strategies, see Table 2.

2.3.2. Working memory: Backward Corsi-like task (Kubota et al., 2020)

In the backward Corsi-like task, each trial started with the presentation of a stimulus (a fairy) in the upper left corner of the display for 1000 ms. Then, the stimulus appeared in random locations within a 3×3 grid of grey squares (see Fig. 1). Children were to recall the presented sequence in reversed order by tapping the squares the stimulus had appeared in. The stimulus was presented for 1000 ms at each location of a sequence within the grid. There was no inter stimulus interval. Response registration started immediately after the presentation of a sequence and was indicated by the squares of the grid turning white. Once a square was tapped, it turned grey for 200 ms and response registration was stopped for the duration of this interval. A practice phase preceded the test phase including two trials with a sequence length of one and two trials with a sequence length of two. In the practice phase, it was indicated whether the sequence was recalled correctly or incorrectly by images of a green or red star, respectively. Practice trials were repeated until the respective sequence was recalled correctly. The test phase started with a sequence length of one and difficulty was increased up to a maximum of eight squares per sequence. After six trials of one sequence length, it was indicated by an audio track that the sequence length would increase by one location. In addition, the task instruction to recall the sequence in reverse order was repeated. The task ended if children failed to recall at least four out of six trials of a given sequence length. The dependent variable was the product score (sequence length $\times$ number of correctly recalled sequences) which is a more reliable index than maximum span length (Kessels et al., 2000; Kubota et al., 2020). The Spearman-Brown corrected split-half reliability estimates were $r = .91$ in the 5- to 6-year-old and $r = .94$ in the 8- to 12-year-old children.

2.3.3. Working memory: Backward verbal span task (van de Weijer-Bergsma et al., 2016)

In the backward verbal span task, each trial started with the presentation of a fixation cross at the center of the display for 1000 ms. Then, images of stimuli appeared successively at the center of the display for 1000 ms each. After presentation of the last stimulus of a sequence, all stimuli of the respective sequence were presented at random locations of a 3×3 grid. Children were to recall the presented sequence in reversed order by tapping the stimuli displayed within the grid. Response registration started immediately with onset of the grid. Once a stimulus was tapped, it was highlighted by a light grey frame for 200 ms and response registration was stopped during this interval. The inter-trial interval was 1000 ms. Stimuli were images of 36 nameable objects (23 one- and 13 two-syllable names) presented randomly with the exception that they occurred only once per sequence. A practice phase preceded the test phase including two trials with a sequence length of one and two trials with a sequence length of two. In the practice phase, it was indicated whether the sequence was recalled correctly or incorrectly by images of a green or red star, respectively. Practice trials were repeated until the respective sequence was recalled correctly. The test phase started with a sequence length of one and difficulty was increased up to a maximum of eight stimuli per sequence. The task ended if children failed to recall at least four out of six trials of a given sequence length. The dependent variable was the product score (sequence length $\times$ number of correctly recalled sequences). The Spearman-Brown corrected split-half reliability estimates were $r = .88$ in the 5- to 6-year-old and $r = .91$ in the 8- to 12-year-old children.

2.3.4. Inhibition: Go/no-go task (Johann et al., 2022)

In the Go/no-go task, stimuli were presented in the upper half of the display. A blue button was presented below (see Fig. 1). Three go stimuli (donkey, chicken, pig) and one no-go stimulus (cow) were presented in random order with the exception that no-go stimuli were always succeeded by a go stimulus. Children were to respond as quickly as possible by tapping a blue button on screen when a go stimulus (75 % of trials) was presented and to refrain from responding when a no-go stimulus was presented (25 % of trials). Stimulus presentation was 500 ms. In the first wave of data collection, responses were registered within 1000 ms after stimulus onset in both age groups. To adjust the level of difficulty for the 8- to 12-year-old children due to a skewed distribution, response registration was reduced to 750 ms in this age group after stimulus onset in the second wave of data collection. The response-stimulus interval was 1000 ms. The task started with a practice block à 20 trials which was repeated once if the error rate exceeded 25 % of trials. Then, four test blocks à 20 trials were presented. Audio instructions were repeated at the beginning of each test block. The dependent variable was the proportion score (hits on go trials – false alarms on no-go trials). This score takes into account the accuracy in both go trials and no-go trials reflecting the ability to discriminate between the two. High scores indicate good inhibitory control of a highly prepotent response. The Spearman-Brown corrected split-half reliability estimates were $r = .80$ in the 5- to 6-year-old and $r = .88$ in the 8- to 12-

Table 2
Example scenario and strategies of the SKT-SRL representing metacognition in the action phase

	Preschool version	Elementary school version
Scenario	Luisa is currently practicing a magic trick that she wants to perform for her friends at her birthday party in two weeks' time. She thinks the trick is great and has decided to practice two of the steps required for the trick today. However, she has other important things to do today and realizes that the time is running out.	Luisa is currently studying for an upcoming class test in two weeks' time. She finds the topic interesting and has decided to do two exercises. However, she has other important tasks to complete for school today and realizes that the time is running out.
Useful strategies	She should think about what is most important today and perhaps practice just one step.	She should check from time to time whether she is studying in a way that will help her prepare for the class test.
	She should find out which of the steps she wanted to practice today she is not yet so good at. She should then practice this particularly well.	She should consider which tasks are most important today and possibly only work on one exercise.
		She should find out what she has not yet understood and concentrate on that.
	She should go through with her plan for the magic trick and postpone the other tasks.	She should follow through with her plan for studying for the class test and postpone the other tasks.
Less useful strategies	She shouldn't keep thinking about what she should do and when. That would only confuse her.	She shouldn't think any more about her approach. It would only confuse her.
		She should hurry so that she can complete all the tasks.

Note. A male protagonist named David and the pronouns “he”, “his”, “him” were presented for male participants.



Fig. 1. Examples of EF tasks assessing working memory, inhibition, and cognitive flexibility.

year-old children. To account for the adjusted difficulty levels in the main analyses, the proportion score was standardized by calculating the z -value for each case using the mean and standard deviation of the respective subsample (i.e., 8- to 12-year-old children in the first wave of data collection vs. 8- to 12-year-old children in the second wave of data collection). To attain the same metric of the variable in the 5- to 6-year-old children, the proportion score was also standardized within this age group.

2.3.5. Inhibition: AX-continuous performance task (AX-CPT; Johann et al., 2022)

In the AX-CPT, stimuli were presented at the center of the display. Two buttons were presented below. The button on the right-hand side was green and the button on the left-hand side was red. Each trial started with the presentation of a fixation cross at the center of the display for 1000 ms. Then, children were presented with a cue (A = glasses / B = hat) followed by a probe (X = book / Y = broom). Children were to press a green button when the presentation of glasses was followed by the presentation of a book (AX trial). Otherwise, they were to press the red button (BX, AY, BY trials). Cues were presented for 500 ms, followed by an inter-stimulus interval of 1500 ms. Then, probes were presented. Response registration started with the onset of a probe and was stopped after a maximum of 1000 ms in both age groups in the first wave of data collection. To adjust the level of difficulty for the 5- to 6-year-old children due to a skewed distribution, response registration was increased to 1500 ms in this age group in the second wave of data collection. Probes were presented for the duration of response registration. On each trial, it was indicated whether the response was correct or incorrect by images of a green or red star, respectively. There was no inter-trial interval. The task started with a practice block à 20 trials which was repeated once if the error rate exceeded 25 % of trials. Then two blocks à 30 trials were presented. 70 % were AX-trials, the remaining 30 % were equally distributed across AY, BX, and BY trials. The dependent variable was the proportion score (hits in AX trials – false alarms on AY and BX trials). This score takes into account the accuracy in AX trials as well as AY and BX trials which require high levels of inhibition as both contain stimuli also presented in AX trials. The Spearman-Brown corrected split-half reliability estimates were $r = .74$ in the 5- to 6-year-old and $r = .62$ in the 8- to 12-year-old children. To account for the adjusted difficulty levels in the main analyses, the proportion score was standardized by calculating the z -value for each case using the mean and standard deviation of the respective subsample (i.e., 5- to 6-year-old children in the first wave of data collection vs. 5- to 6-year-old children in the second wave of data collection). To attain the same metric of the variable in the 8- to 12-year-old children, the proportion score was also standardized within this age group.

2.3.6. Cognitive flexibility: Flexible item selection task (FIST; Kuhn et al., 2017)

In the FIST, children were first introduced with the dimensions of color (red, blue, yellow), number (one, two, three), and type of object (fairy, hat, flower) on the task cards. The task consisted of two blocks. In block 1, children were presented with two cards on the left and the center of the display which matched in one of the three dimensions. The matching dimension was made explicit by an audio track. After the audio track had stopped (1000 ms), a third card was presented on the right side of the display alongside the first two cards (see Fig. 1). This card matched one of the first two cards in a different dimension. Children were to identify the card that matched the new one by tapping the respective card on the display. This required switching attention from one dimension to the target dimension. Responses were registered with the onset of the third stimulus card until the child responded or for a maximum of 5000 ms. In block 2, three stimulus cards were presented at once. One of the three cards matched each of the other two in a different dimension, respectively. Children were to identify this card by tapping it on the display. This required the identification of one dimension and switching attention to the second dimension. Responses were registered from stimulus onset until the child responded or for a maximum of 10 000 ms. In block 1 and 2, the inter-trial interval was 1000 ms. Block 1 included 18 trials, block 2 included nine trials. Each block was preceded by three practice trials with visual feedback of a green and red star indicating whether the response was correct or incorrect, respectively. Practice trials were repeated until the response was correct. The dependent variable was the rate of errors in relation to all trials. The Spearman-Brown corrected split-half reliability estimates were $r = .85$ in the 5- to 6-year-old and $r = .79$ in the 8- to 12-year-old children.

2.3.7. Cognitive flexibility: Modified Wisconsin card sorting test (M-WCST; Cianchetti et al., 2007)

In the M-WCST, four stimulus cards displaying symbols differing in color, shape, and number (1. single red triangle, 2. two green stars, 3. three yellow crosses, 4. four blue circles) were presented in the upper half of the display. Children were to sort a set of response cards with the stimulus cards according to one of the dimensions of color, shape, or number. After six consecutive sorts along one dimension (i.e., one category), the requirement to switch the dimension was indicated by an audio track and a visual cue. Now,

children were to identify a different dimension than the previous one and sort the response cards with the stimulus cards accordingly for six consecutive trials. The task continued until children had completed six consecutive sorts of each dimension in each of two test blocks or until the maximum of 48 response cards had been presented. The task started with a practice block à 12 trials. In the practice trials, response cards were presented until the child responded. Practice trials were repeated once if the response was incorrect. Then, two test blocks à 18–24 trials (depending on performance) followed. In the test trials, response cards were presented until the child responded or up to a maximum of 8000 ms. There was no inter-trial interval. On each trial, it was indicated whether the response was correct or incorrect by images of a green or red star, respectively. The dependent variable was the rate of errors in relation to number of trials. Trials were considered erroneous when a) children failed to sort according to a new dimension after the indication to switch dimensions, b) children failed to sort according to a new dimension despite negative feedback on the previous trial indicating that the sorting dimension was incorrect, and c) children failed to sort according to the current category (i.e., sorting according to a different dimension although one category of six consecutive sorts of the current dimension was not yet completed). The Spearman-Brown corrected split-half reliability estimates were $r = .87$ in the 5- to 6-year-old and $r = .73$ in the 8- to 12-year-old children.

2.3.8. Fluid intelligence: Matrix reasoning subtest of the Kaufman assessment battery for children – second edition (K-ABC-II; Kaufman & Kaufman, 2015)

On the matrix reasoning subtest of the K-ABC-II, children were presented with a series of five figurative elements and one element with a question mark. They were to replace the element with the question mark by selecting one of four to six figurative response alternatives to match the series. The test started with one introductory trial. Then, an age-adjusted number of trials was presented with a maximum of 21 trials in 5–6-year-old and 30 trials in 8–12-year-old children. The sum of correctly solved trials was calculated separately for odd and even items to create two dependent variables. This allowed for modeling fluid intelligence as a latent variable in the main analyses, controlling for task-specific and error variance. Cronbach's α of the scales was .80 and .79 in the 5- to 6-year-old and .79 in the 8- to 12-year-old children.

2.4. Data analyses

All analyses were conducted with RStudio (Posit Team, 2024). Descriptive statistics were computed with the psych package (Revelle, 2023). On the executive function measures and the SKT-SRL, children who had completed < 50 % of all items/trials due to premature task abortion or showed response bias were coded missing on the respective task. This ranged between 0.41 % and 3.6 % of cases. These children refrained from responding on the majority of trials or responded equally to the majority of go- and no-go trials on the Go/no-go task, indicating that they did not understand the differential rules of the task. On the SKT-SRL, response bias was indicated when children rated the majority of strategies with the same number stars. In line with the test manual of the K-ABC-II (Kaufman & Kaufman, 2015), children who scored 0 on the first three trials of the matrix reasoning subtest were coded missing (1.32 % in preschool). On EF tasks, values above or below 3 SD of the between-subjects mean were replaced by the most extreme value within 3 SD of the distribution (Kline, 2016). This ranged between 0.4 % and 2.4 % of cases. On the SKT-SRL, an influential datapoint was identified in the 8- to 12-year-old children and was coded missing on all SRL variables. We estimated a linear model regressing the metacognition-reflection scale onto the remaining scales of the SKT-SRL and conducted multivariate outlier analysis. The influential datapoint was identified by plotting the standardized residuals against leverage. Practice blocks were not analyzed, neither were the first trial/block and trials below 200 ms on the Go/no-go task and AX-CPT. On the M-WCST, a technical problem resulted in a loss of data in block 2 of 51 children (10.85 %). Thus, only block 1 was retained in the analyses for those children. Error rates of the M-WCST and FIST were recoded so that higher values indicated better performance.

The shared and unique contributions of executive functions and fluid intelligence to conditional strategy knowledge of SRL were analyzed by means of SEM using the lavaan package (version: 0.6–17; Rosseel, 2012). Full information maximum likelihood (FIML) estimation was applied to account for missing data. Due to skewness and/or kurtosis of the variable distributions, robust standard errors were estimated (MLR). One factor loading for each latent factor was fixed to one for model identification. Following Beauducel and Wittmann (2005), several fit indices were considered to evaluate model fit. These included the Satorra-Bentler scaled $\chi^2(df)$ and its p -value, the Comparative Fit Index (CFI), the Root Mean Square Error of Approximation (RMSEA), and the Standardized Root Mean Square Residual (SRMR). Good model fit was indicated by a non-significant χ^2_{SB} -value, $CFI \geq .95$, $RMSEA \leq .05$, and $SRMR \leq .06$.

3. Results

3.1. Descriptive statistics

Univariate descriptive statistics of executive function, fluid intelligence, and SKT-SRL variables are displayed in Table 3 for each age group. Spearman-Rho correlations between manifest executive function, fluid intelligence, and SKT-SRL variables are displayed in Table 4 for each age group.

3.2. Contributions of executive functions and fluid intelligence to conditional strategy knowledge of SRL in 5- to 6-year-old and 8- to 12-year-old children

Previous analyses on the factorial structure of executive functions (Grobe et al., 2024) as well as conditional strategy knowledge of SRL in subgroups of the sample under study revealed that a one-factor model for both constructs provided the best fit with the data in

Table 3

Descriptive statistics of executive functions, fluid intelligence and conditional strategy knowledge of SRL by age group

Measures	5- to 6-year-old children						8- to 12-year-old children					
	<i>n</i>	<i>M</i>	<i>SD</i>	Range	Skew	Kurtosis	<i>n</i>	<i>M</i>	<i>SD</i>	Range	Skew	Kurtosis
Working memory: Verbal span task	250	24.11	13.54	6–69	1.28	1.69	199	49.47	21.70	6–116.80	0.30	0.01
Working memory: Corsi-like task	252	19.78	11.90	0–56.97	0.79	0.13	209	61.86	27.66	6–130	0.08	–0.47
Inhibition: AX-CPT ^a	200	0.00	1.00	–2.63–2.13	–0.51	–0.17	198	0.01	0.98	–2.99–1.07	–1.19	0.89
Inhibition: Go/no-go task ^a	250	0.00	1.00	–2.99–2.13	–0.60	0.25	210	0.00	1.00	–2.97–1.99	–0.76	–0.05
Cognitive flexibility: M-WCST ^b	223	60.50	14.99	14.88–89.58	–0.73	0.41	200	77.63	12.63	37.78–100	–1.01	1.33
Cognitive flexibility: FIST ^b	244	63.75	17.45	25.93–100	0.19	–0.78	208	91.74	9.48	52.83–100	–1.26	1.32
Intelligence: K-ABC-II (even)	224	3.39	2.35	0–10	0.00	–0.83	207	10.00	2.84	0–15	–0.57	0.03
Intelligence: K-ABC-II (odd)	224	3.83	2.24	0–10	0.36	–0.52	207	10.73	2.82	1–15	–0.67	–0.10
SKT-SRL: Metacognition/Planning	240	3.50	2.62	0–8	0.24	–1.06	207	9.44	4.54	0–18	0.02	–0.73
SKT-SRL: Metacognition/Action	241	3.49	2.59	0–8	0.23	–0.97	206	8.77	4.60	0–18	0.18	–0.73
SKT-SRL: Metacognition/Reflection	234	2.14	2.15	0–8	0.83	–0.04	205	6.47	4.11	0–17	0.32	–0.81
SKT-SRL: Motivation/Planning	240	4.90	2.51	0–8	–0.28	–1.03	207	9.58	4.11	2–18	0.01	–0.86
SKT-SRL: Motivation/Action	238	4.55	2.80	0–8	–0.23	–1.23	206	8.75	4.11	0–18	–0.08	–0.71
SKT-SRL: Motivation/Reflection	236	2.39	2.29	0–8	0.65	–0.51	207	8.10	3.81	0–18	0.21	–0.40
SKT-SRL: Cognition/Action	236	2.93	2.52	0–8	0.46	–0.81	208	7.55	3.42	0–17	0.02	–0.50

Note. AX-CPT: AX – continuous performance task, FIST: Flexible item selection task, M-WCST: Modified Wisconsin card sorting test, K-ABC-II: Matrix reasoning subtest of the Kaufman assessment battery for children – second edition, SKT-SRL: Strategy knowledge test for self-regulated learning.

^a Standardized values were used for the AX-CPT and Go/no-go proportion scores.

^b Dependent variables of the FIST and M-WCST are inversely coded (i.e., larger values represent better performance).

Table 4

Spearman-Rho correlations between measures of executive functions, fluid intelligence and conditional strategy knowledge of SRL by age group

Measures	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.
1. Working memory: Verbal span task	-	.28**	.24**	.19*	.26**	.27**	.30**	.26**	.15*	.14	-.09	.06	.09	.07	.09
2. Working memory: Corsi-like task	.52**	-	.20**	.09	.20*	.28**	.32**	.33**	.08	.02	-.09	.06	-.02	.06	.03
3. Inhibition: AX-CPT ^a	.37**	.29**	-	.31**	.10	.10	.13	.12	.00	.11	-.10	.16*	.11	.09*	.21**
4. Inhibition: Go/no-go ^a	.27**	.19**	.23**	-	-.05	.11	.05	.04	.03	.02	-.01	-.04	.21**	.03	.10
5. Cognitive flexibility: M-WCST ^b	.12	.16*	.12	.09	-	.21**	.25**	.30**	.01	.08	-.13	.18*	-.02	.00	.00
6. Cognitive flexibility: FIST ^b	.38**	.34**	.33**	.15*	.16*	-	.28**	.33**	.09	.05	-.06	.03	.08	-.10	.14*
7. Intelligence: K-ABC-II (even)	.37**	.34**	.22**	.15*	.25**	.27**	-	.82**	.00	.05	-.12	.09	.04	-.01	.15*
8. Intelligence: K-ABC-II (odd)	.42**	.37**	.25**	.18*	.22**	.32**	.86**	-	-.07	.02	-.08	.04	.01	-.05	.13
9. SKT-SRL: Metacognition/Planning	.11	.02	.04	.08	.00	.11	.05	.11	-	.25**	.19*	.11	.13	.09	.13
10. SKT-SRL: Metacognition/Action	.07	.04	.09	.14*	.04	.11	-.08	-.07	.19**	-	.10	.22**	.20**	.13	.29**
11. SKT-SRL: Metacognition/Reflection	-.08	.00	.03	.14*	.02	.05	.06	.04	.11	.08	-	-.02	.10	.08	.11
12. SKT-SRL: Motivation/Planning	.18**	.18*	.14	.14*	.07	.22**	.23**	.21**	.22**	.22**	.01	-	.09	.10	.13
13. SKT-SRL: Motivation/Action	.20**	.01	.02	.15*	-.03	.12	.09	.12	.28**	.12	.05	.16*	-	.22**	.09
14. SKT-SRL: Motivation/Reflection	.03	-.09	.06	.12	-.05	.01	-.07	-.03	.12	.13*	.23**	.02	.16*	-	.17*
15. SKT-SRL: Cognition/Action	.10	.01	-.02	.03	-.08	.12	.04	.03	.16*	.16*	.21**	.08	.22**	.13*	-

Note. Correlation coefficients for the 5- to 6-year-old children below the diagonal, for the 8- to 12-year-old children above the diagonal. AX-CPT: AX – continuous performance task, FIST: Flexible item selection task, M-WCST: Modified Wisconsin card sorting test, K-ABC-II: Matrix reasoning subtest of the Kaufman assessment battery for children – second edition, SKT-SRL: Strategy knowledge test for self-regulated learning.

^a Standardized values were used for the AX-CPT and Go/no-go proportion scores.

^b Dependent variables of the FIST and M-WCST are inversely coded (i.e., larger values represent better performance).

* $p < .05$,

** $p < .01$

both the 5- to 6-year-old and 8- to 12-year-old children, respectively (see Appendix A for supplementary analyses). Therefore, we adopted the one-factor solution for executive functions with six indicators loading on a common factor in each age group (with one error covariance in one age group). Likewise, a one-factor model with seven indicators from the SKT-SRL was adopted for conditional strategy knowledge of SRL in each age group (also with one error covariance in one age group).

Then, a model with executive functions and fluid intelligence as latent predictor variables and conditional strategy knowledge of SRL as a latent criterion variable was estimated in each age group. For the 5- to 6-year-old children (see Fig. 2), the model provided very good fit with the data ($\chi^2_{SB}(df) = 93.44$ (86), $p = .27$, CFI = .99, RMSEA [90 % CI] = .02 [.01–.04], SRMR = .06). Executive functions significantly predicted conditional strategy knowledge of SRL ($\beta = .36$, $p = .02$), whereas fluid intelligence was not predictive of conditional strategy knowledge of SRL ($\beta = -.04$, $p = .79$). The latent correlation between executive functions and fluid intelligence was significant ($r = .60$, $p < .001$). When executive functions were not controlled for, fluid intelligence and conditional strategy knowledge of SRL were unrelated as well ($r = .18$, $p = .09$). The total amount of variance explained in conditional strategy knowledge of SRL was 11.6 % ($R^2 = .116$).

For the 8- to 12-year-old children (see Fig. 3), the model fit was good ($\chi^2_{SB}(df) = 105.57$ (86), $p = .08$, CFI = .96, RMSEA [90 % CI] = .03 [.01–.05], SRMR = .06). Executive functions significantly predicted conditional strategy knowledge of SRL ($\beta = .41$, $p = .04$), but fluid intelligence did not ($\beta = -.17$, $p = .36$). The latent correlation between executive functions and fluid intelligence was significant ($r = .60$, $p < .001$). When executive functions were not controlled for, fluid intelligence and conditional strategy knowledge of SRL were unrelated as well ($r = .11$, $p = .32$). The total amount of variance explained in conditional strategy knowledge of SRL was 11.5 % ($R^2 = .115$).

Model comparisons support a one-dimensional EF structure in the current sample (Grobe et al., 2024), however, supplementary analyses were conducted to gain insight into the relations between executive function components and conditional strategy knowledge of SRL. Specifically, a model with latent factors for working memory, inhibition, cognitive flexibility, fluid intelligence, and conditional strategy knowledge of SRL was estimated in each age group and latent correlations were freely estimated. Regarding the 5- to 6-year-old children, see Fig. 4. Regarding the 8- to 12-year-old children, see Fig. 5.

4. Discussion

The present study aimed at examining the shared and unique contributions of executive functions and fluid intelligence to conditional strategy knowledge of SRL in a sample of 5- to 6-year-old and 8- to 12-year-old children by means of latent variable analysis. As

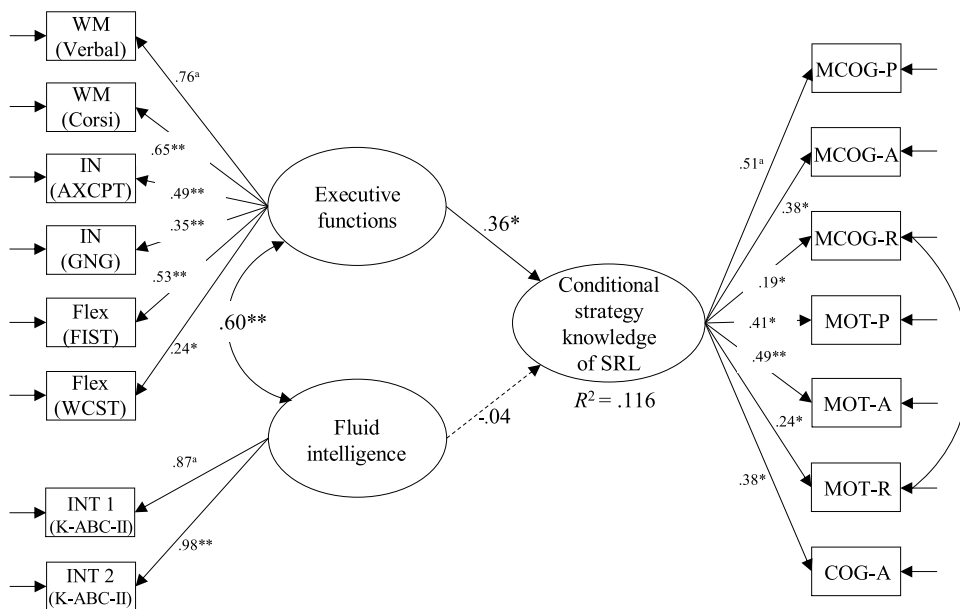


Fig. 2. SEM with executive functions and fluid intelligence as predictors of conditional strategy knowledge of SRL in the 5- to 6-year-old children. The fit of the model to the data was very good: $\chi^2_{SB}(df) = 93.44$ (86), $p = .27$, CFI = .99, RMSEA [90 % CI] = .02 [.01–.04], SRMR = .06. Significant paths and factor loadings are indicated by solid arrows. WM: Working memory, IN: Inhibition, Flex: Cognitive flexibility, INT: Fluid intelligence, Corsi: Backward Corsi-like task, VS: Backward verbal span task, FIST: Flexible item selection task, WCST: Modified Wisconsin card sorting test, GNG: Go/no-go task, AX-CPT: AX – continuous performance task, K-ABC-II: Matrix reasoning subtest of the Kaufman assessment battery for children – second edition, MCOG-P: Metacognition/Planning scale of the SKT-SRL, MCOG-A: Metacognition/Action scale of the SKT-SRL, MCOG-R: Metacognition/Reflection scale of the SKT-SRL, MOT-P: Motivation/Planning scale of the SKT-SRL, MOT-A: Motivation/Action scale of the SKT-SRL, MOT-R: Motivation/Reflection scale of the SKT-SRL, COG: Cognition/Action scale of the SKT-SRL. ^a p -values were estimated based on unstandardized parameters. The first unstandardized loading of each latent variable was fixed to one. Standardized parameters are depicted. * $p < .05$, ** $p < .01$.

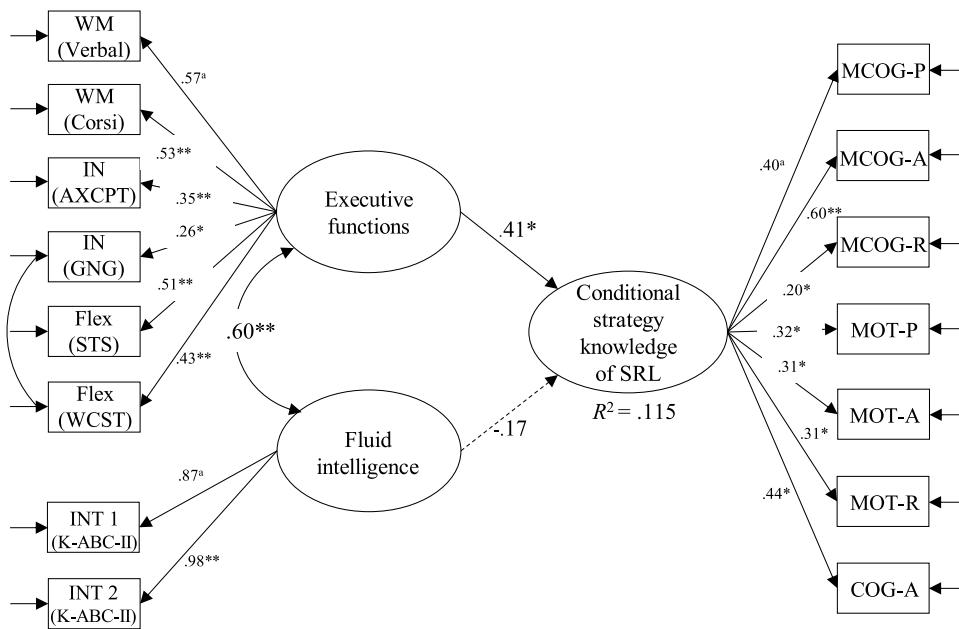


Fig. 3. SEM with executive functions and fluid intelligence as predictors of conditional strategy knowledge of SRL in the 8- to 12-year-old children. The fit of the model to the data was good: $\chi^2_{df} = 105.57$ (86), $p = .08$, CFI = .96, RMSEA [90 % CI] = .03 [.01–.05], SRMR = .06. Significant paths and factor loadings are indicated by solid arrows. WM: Working memory, IN: Inhibition, Flex: Cognitive flexibility, INT: Fluid intelligence, Corsi: Backward Corsi-like task, VS: Backward verbal span task, FIST: Flexible item selection task, WCST: Modified Wisconsin card sorting test, GNG: Go/no-go task, AX-CPT: AX – continuous performance task, K-ABC-II: Matrix reasoning subtest of the Kaufman assessment battery for children – second edition, MCOG-P: Metacognition/Planning scale of the SKT-SRL, MCOG-A: Metacognition/Action scale of the SKT-SRL, MCOG-R: Metacognition/Reflection scale of the SKT-SRL, MOT-P: Motivation/Planning scale of the SKT-SRL, MOT-A: Motivation/Action scale of the SKT-SRL, MOT-R: Motivation/Reflection scale of the SKT-SRL, COG: Cognition/Action scale of the SKT-SRL. ^a p -values were estimated based on unstandardized parameters. The first unstandardized loading of each latent variable was fixed to one. Standardized parameters are depicted. * $p < .05$, ** $p < .01$.

expected, the latent predictors of executive functions and fluid intelligence were significantly correlated suggesting strong overlap of these higher-order cognitive abilities during childhood. In line with our expectations, executive functions significantly predicted conditional strategy knowledge of SRL in both 5- to 6-year-old and 8- to 12-year-old children. In contrast, exploratory analyses on the predictive role of fluid intelligence for conditional strategy knowledge of SRL revealed that fluid intelligence did not significantly contribute to the prediction in either age group.

4.1. Prediction of conditional strategy knowledge of SRL from executive functions

With regard to the prediction of conditional strategy knowledge of SRL from executive functions in both the 5- to 6-year-old and 8- to 12-year-old children, our findings provide empirical evidence for theoretical claims on a conceptual link between executive functions and SRL competencies (Hoyle & Dent, 2018; Pintrich & Zusho, 2002). The results of the present study indicate associations of moderate strength in both the 5- to 6-year-old and 8- to 12-year-old children. However, supplementary analyses suggest differential relations between executive function components and conditional strategy knowledge of SRL across age groups.

The significant contribution of executive functions to conditional strategy knowledge of SRL in the 5- to 6-year-old children complements previous findings on positive associations of executive functions and enacted SRL in children of similar age (Davis et al., 2021). Davis et al. (2021) reported that a common factor model in which executive function and SRL measures loaded on the same latent variable provided very good fit with the data and concluded that executive function performance and SRL might be manifestations of the same underlying self-regulation construct in preschool and kindergarten children.

In the present study, the association between executive functions and conditional strategy knowledge of SRL in the 5- to 6-year-old children does not suggest a strong overlap of the constructs as reported by Davis et al. (2021). These authors operationalized SRL in terms of the frequency of classroom behaviors related to the cognitive, metacognitive, and motivational components of SRL via teacher reports. The present study assessed conditional strategy knowledge of SRL, that is children's knowledge of when and why to apply motivational, metacognitive, or cognitive strategies given a specific learning problem in the three learning phases (Paris et al., 1983). Potentially, enacting cognitive, metacognitive, and motivational behaviors related to SRL in the real-life context of a classroom implicates children's executive functions more strongly than the activation of conditional strategy knowledge of SRL on a scenario-based test. This might explain higher associations between measures of executive functions and strategy use. In fact, the execution of learning strategies is mentally effortful and draws on large proportions of cognitive capacity, especially in young children (Bjorklund et al.,

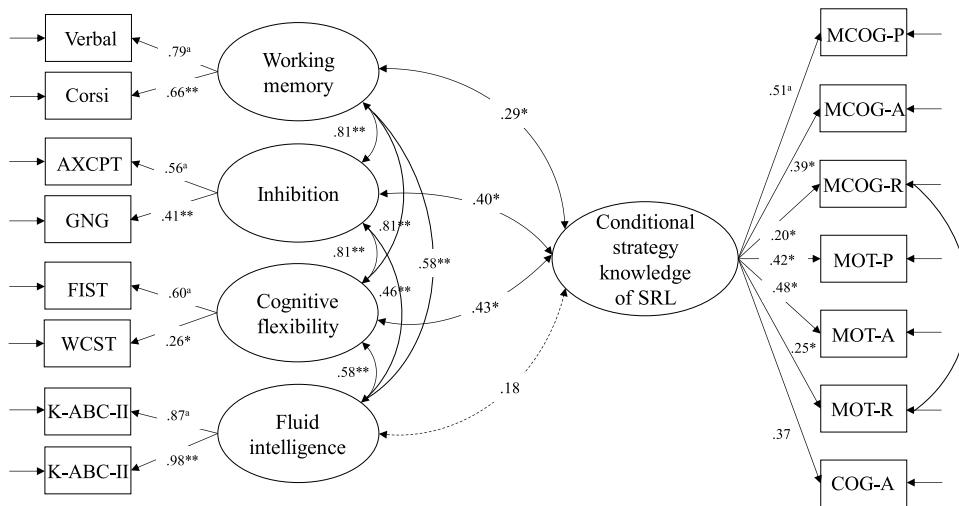


Fig. 4. SEM with latent correlations between working memory, inhibition, cognitive flexibility, fluid intelligence, and conditional strategy knowledge of SRL in the 5- to 6-year-old children. The fit of the model to the data was very good: $\chi^2_{SB}(df) = 87.15$ (79), $p = .25$, CFI = .99, RMSEA [90 % CI] = .02 [.01–.04], SRMR = .05. Significant correlations and factor loadings are indicated by solid arrows. Corsi: Backward Corsi-like task, Verbal: Backward verbal span task, GNG: Go/no-go task, AXCPT: AX – continuous performance task, FIST: Flexible item selection task, WCST: Modified Wisconsin card sorting test, K-ABC-II: Matrix reasoning subtest of the Kaufman assessment battery for children – second edition, MCOG-P: Metacognition/Planning scale of the SKT-SRL, MCOG-A: Metacognition/Action scale of the SKT-SRL, MCOG-R: Metacognition/Reflection scale of the SKT-SRL, MOT-P: Motivation/Planning scale of the SKT-SRL, MOT-A: Motivation/Action scale of the SKT-SRL, MOT-R: Motivation/Reflection scale of the SKT-SRL, COG: Cognition/Action scale of the SKT-SRL. ^a p -values were estimated based on unstandardized parameters. The first unstandardized loading of each latent variable was fixed to one. Standardized parameters are depicted. * $p < .05$, ** $p < .01$.

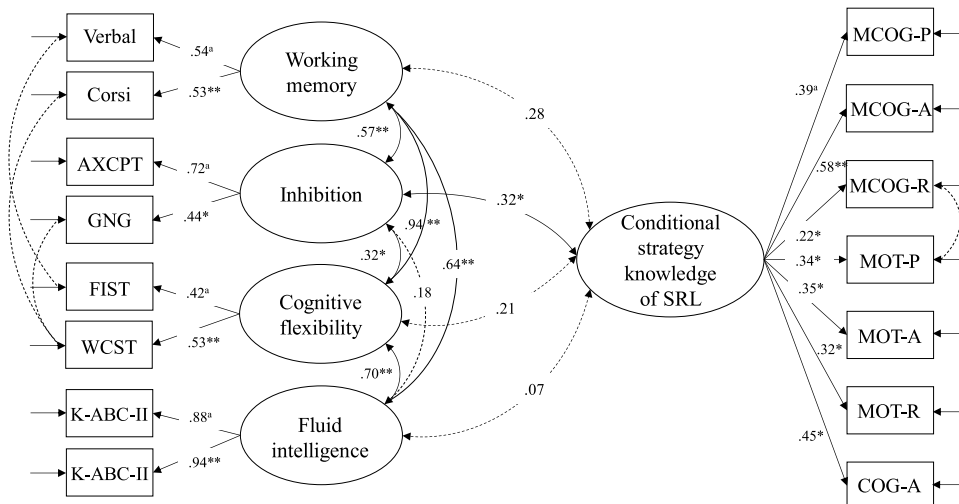


Fig. 5. SEM with latent correlations between working memory, inhibition, cognitive flexibility, fluid intelligence, and conditional strategy knowledge of SRL in the 8- to 12-Year-old children. The fit of the model to the data was very good: $\chi^2_{SB}(df) = 81.96$ (76), $p = .30$, CFI = .99, RMSEA [90 % CI] = .02 [.01–.05], SRMR = .05. Error covariances were added to avoid an out of range factor correlation. Significant correlations and factor loadings are indicated by solid arrows. Corsi: Backward Corsi-like task, Verbal: Backward verbal span task, GNG: Go/no-go task, AXCPT: AX – continuous performance task, FIST: Flexible item selection task, WCST: Modified Wisconsin card sorting test, K-ABC-II: Matrix reasoning subtest of the Kaufman assessment battery for children – second edition, MCOG-P: Metacognition/Planning scale of the SKT-SRL, MCOG-A: Metacognition/Action scale of the SKT-SRL, MCOG-R: Metacognition/Reflection scale of the SKT-SRL, MOT-P: Motivation/Planning scale of the SKT-SRL, MOT-A: Motivation/Action scale of the SKT-SRL, MOT-R: Motivation/Reflection scale of the SKT-SRL, COG: Cognition/Action scale of the SKT-SRL. ^a p -values were estimated based on unstandardized parameters. The first unstandardized loading of each latent variable was fixed to one. Standardized parameters are depicted. * $p < .05$, ** $p < .01$.

2009; Winne, 2011). However, our findings are in line with previous studies suggesting that the role of executive functions is not limited to supporting the execution of cognitive, metacognitive, or motivational strategies (e.g., Geurten et al., 2016). More specifically, the contribution of executive functions to conditional strategy knowledge of SRL may be interpreted in the sense that executive functions aid in the identification of adequate cognitive, metacognitive, or motivational strategies when facing a specific learning problem. This interpretation corresponds with previous results on the association between executive functions and the identification and choice of learning goals in preschool-aged children stressing the importance of executive functions for the planning of learning actions (Leclercq et al., 2023). The results of the present study add to these findings. Specifically, executive functions might support the activation of strategy knowledge that is relevant for working on a learning task and goal achievement (Boekaerts, 1999; Winne, 2011; Zimmerman, 2000). Thus, executive functions might not only be involved in the overt application of learning strategies (e.g., Davis et al., 2021), but also in more covert processes of SRL such as goal identification and knowledge activation.

The significant contribution of executive functions to conditional strategy knowledge of SRL in the 8- to 12-year-old children aligns with previous results on positive links between executive functions and enacted SRL in children of similar age (Ferreira et al., 2022; Rutherford et al., 2018). As is the case for studies in preschool samples, methodological differences between studies in elementary school samples lead to somewhat different interpretations of the results. Since we operationalized SRL in terms of conditional strategy knowledge as opposed to SRL in the classroom (Rutherford et al., 2018), the same interpretation as formulated for the findings in the 5- to 6-year-old children might apply to the 8- to 12-year-olds in the present study. We can only conclude that executive functions contribute to better identification of useful cognitive, metacognitive, and motivational strategies in specific learning situations, not actual strategy usage. Another methodological difference in previous studies in this age group concerns the conceptualization of executive functions. By analyzing the unique contributions of working memory, inhibition, and cognitive flexibility to teacher-reported SRL, Rutherford et al. (2018) were able to distinguish which of the core executive functions were involved in SRL. They found that both inhibition and cognitive flexibility significantly predicted SRL, whereas working memory did not. Other studies did find that working-memory performance significantly predicted SRL (Ferreira et al., 2022). In the present study, executive function performance was represented by a common factor in both age groups. However, supplementary analyses revealed that inhibition was driving the association between executive functions and conditional strategy knowledge of SRL in the 8- to 12-year-old children, whereas all executive function components were involved in the 5- to 6-year-old children. This finding suggests age-related changes in the role of specific executive functions for conditional strategy knowledge of SRL. Potentially, as children gain experience with learning strategies and their conditional strategy knowledge of SRL becomes more routinely activated at the end of elementary school, its activation might not require working memory and cognitive flexibility to the same extent as in earlier developmental stages. Rather, inhibition might be decisive in suppressing less useful strategies in favor of useful ones given a specific learning situation. While these supplemental analyses provide insight into the differential roles of executive function components for conditional strategy knowledge of SRL, it must be acknowledged, that modeling executive functions as a multidimensional construct with three correlated factors came with challenges. Specifically, in the 8- to 12-year-olds, error covariances were added to avoid out of range correlations between factors suggesting that executive functions were still closely entangled in this age group. The finding that only inhibition was significantly related to conditional strategy knowledge of SRL in this age group should therefore be considered preliminary. Moreover, contributions of executive functions to single components or phases proposed in theoretical models of SRL (e.g., Boekaerts, 1999; Zimmerman, 2000; Zimmerman & Moylan, 2009) warrant further research as conditional strategy knowledge of SRL is only an aspect of the broad SRL construct and was conceptualized as a unidimensional factor in the present study.

The distinction between conditional strategy knowledge and strategy usage of SRL closely resembles that of metacognitive knowledge and metacognitive skills incorporated in the construct of metacognition (Efklides, 2008) which is integral in theoretical models of SRL (e.g., Efklides, 2011; Efklides & Schwartz, 2024). From a theoretical standpoint, executive functions and metacognition are also closely related (Roebbers, 2017), and empirical studies in preschool- and elementary school-aged children report on significant links between these constructs (Bryce et al., 2015; Roebbers et al., 2012). Studies in adolescents and adults do suggest a mediating role of metacognition in the relation between executive functions and enacted SRL (Effeney et al., 2013; Follmer & Sperling, 2016). Therefore, it would be useful to integrate metacognition in future studies with childhood samples in order to shed light on the mechanism linking executive functions and conditional knowledge about, as well as the usage of cognitive, metacognitive, and motivational strategies.

4.2. Prediction of conditional strategy knowledge of SRL from fluid intelligence

Although significantly correlated with executive functions in both age groups, fluid intelligence did not significantly contribute to the prediction of conditional strategy knowledge of SRL in either age group in the present study. With regard to enacted SRL, our findings contrast with studies in the field that reported an advantage in strategy use in highly intelligent students (e.g., Bouffard-Bouchard et al., 1993; Zimmerman & Martinez-Pons, 1990). However, findings of the present study are hardly comparable to these studies due to higher age means in their samples and different methodological approaches. Early studies on the relation of intelligence and SRL have treated intelligence as a categorical variable and compared highly intelligent and average-intelligent youths in their application of cognitive, metacognitive, and motivational strategies (Bouffard-Bouchard et al., 1993; Zimmerman & Martinez-Pons, 1990). This practice discards individual differences in intelligence and can lead to a loss of statistical power and increased Type I error rates (MacCallum et al., 2002). Potentially, this methodological shortcoming may be one reason for the rather inconsistent results on the relations between intelligence and SRL (Dresel & Haugwitz, 2005). When intelligence was treated as a continuous variable, null or negative associations between intelligence and measures of SRL in school-aged children were obtained in previous studies (Dresel & Haugwitz, 2005; Sontag et al., 2012). Our findings corroborate and add to these results as they indicate that intelligence does not

contribute significantly to conditional strategy knowledge of SRL in both the 5- to 6-year-old and 8- to 12-year-old children.

With regard to studies on the relation of intelligence and metacognitive knowledge, the results of the present study contrast with two early studies in the field that proposed an advantage of highly intelligent children in metacognitive knowledge. Specifically, our results obtained in the 5- to 6-year-old children do not align with findings by [Schwanenflugel et al. \(1997\)](#), who found that highly intelligent kindergarten and first-grade children were more knowledgeable of memory and attention processes than average-intelligent peers. Yet, the authors stated that this difference was quite small. Similarly, the findings obtained in the 8- to 12-year-old children are not in line with previous studies in this age group reporting on positive links between intelligence and metacognitive knowledge about the usefulness of a sorting strategy in a sort-recall task ([J.M. Alexander & Schwanenflugel, 1994](#)). One explanation for these differences might be the different operationalizations of the constructs under study. We operationalized intelligence in terms of reasoning ability, a facet of fluid intelligence ([Cattell, 1963](#)), whereas previous investigations assessed both non-verbal and verbal aspects of intelligence ([J.M. Alexander & Schwanenflugel, 1994](#); [Schwanenflugel et al., 1997](#)). It is possible that verbal abilities differentially contribute to SRL competencies in childhood ([Jacob et al., 2019](#)), but were not captured by the intelligence measure in the present study. Moreover, the operationalization of conditional strategy knowledge of motivation, metacognition, and cognition within the three learning phases in the present study was different to the conceptualization of metacognitive knowledge in previous studies, which mainly included knowledge about cognitive or attention processes. Small sample sizes ([Schwanenflugel et al., 1997](#)) or highly specific tasks ([J.M. Alexander & Schwanenflugel, 1994](#)) further impede cross-study comparison. Thus, we can only conclude that fluid intelligence as a specific facet of the broad intelligence construct on average does not contribute to conditional strategy knowledge of SRL in children. The predictive role of other facets of intelligence for conditional strategy knowledge of SRL and other aspects of the complex SRL construct remains to be analyzed.

4.3. Limitations and implications for future research

The current study analyzed the relations between executive functions, fluid intelligence, and conditional strategy knowledge of SRL by means of a latent variable approach in two groups of preschool and fourth-grade elementary school children aged 5–6 and 8–12 years, respectively. This allowed for the control of measurement error resulting in more accurate estimations of the contributions from executive functions and fluid intelligence to conditional strategy knowledge of SRL. Moreover, the comparative approach helped to shed light on differential relations among these constructs and potential differences between age groups.

Nevertheless, there are limitations in the current study that should be discussed. First, while we aimed to reach high comparability by using the same executive function and fluid intelligence measures and age-adapted versions of the SKT-SRL across age groups, this approach came along with methodological challenges. Changes to the Go/no-go task in the 8- to 12-year-old children during the study might have caused systematic error covariance between the Go/no-go task and M-WCST. Another explanation includes the notion that inhibition was starting to separate from a common executive function factor in middle childhood (e.g., [Brydges et al., 2014](#)) and performance on the Go/no-go task assessing inhibition was therefore not fully represented by the common factor. With regard to conditional strategy knowledge of SRL, we allowed an error covariance between the indicators of metacognitive and motivational strategy knowledge in the reflection phase in the 5- to 6-year-old children. Young children possess only limited cognitive abilities necessary to reflect on learning behavior ([Roebbers, 2014](#); [Schneider, 2008](#)). The limited ability to reflect likely affects preschool children's knowledge regarding the usefulness of strategies supporting reflection processes. This potentially affected both scales assessing conditional strategy knowledge in the reflection phase explaining systematic error variance between these two indicators.

Moreover, different item content in the preschool and elementary school versions of the SKT-SRL precluded tests of measurement invariance across age groups. The interpretations regarding similarities and differences in the prediction of conditional strategy knowledge of SRL from executive functions and fluid intelligence across age groups must therefore be considered preliminary.

Second, the current study applied a scenario-based test of conditional strategy knowledge ([Dörrenbächer-Ulrich et al., 2024](#)) which is considered a central competence and prerequisite for the application of cognitive, metacognitive, and motivational strategies ([Dresel et al., 2015](#); [Karlen, 2015](#)). However, this relation has not consistently been found empirically ([Dörrenbächer-Ulrich et al., 2021](#)). Therefore, we cannot draw any conclusion on how executive functions and fluid intelligence contribute to the use of cognitive, metacognitive, and motivational strategies in children based on the current findings. Different authors have recommended the multi-method assessment of SRL to adequately account for the complex nature of the construct ([Ben-Eliyahu & Bernacki, 2015](#); [Cleary & Callan, 2018](#)). This would also help to shed light on the relation between conditional strategy knowledge of SRL and enacted SRL. Accordingly, future studies in children should combine instruments targeting conditional strategy knowledge of SRL as well as actual strategy usage, for example teacher-report questionnaires (see [Davis et al., 2021](#); [Rutherford et al., 2018](#)) or micro-analytic protocols as a process-oriented, online assessment method of SRL ([Cleary et al., 2012](#); [Wirth & Leutner, 2008](#)).

Third, fluid intelligence was assessed with only one measure in the current study. Specifically, we assessed reasoning ability reflecting fluid intelligence, which is the ability to adapt and solve problems in novel situations ([Cattell, 1963](#)). However, intelligence is a multifaceted construct encompassing also the ability to use experience and acquired skills, namely crystallized intelligence ([Cattell, 1963](#)). Aspects of crystallized intelligence, such as verbal ability, have been proposed to contribute to the development of SRL in childhood ([Jacob et al., 2019](#); [Hoyle & Dent, 2018](#)). Thus, the inclusion of measures assessing crystallized intelligence can help to gain more clarity on the association of intelligence and SRL in children.

Fourth, while the current study revealed that executive functions significantly predicted conditional strategy knowledge of SRL in both the 5- to 6-year-old and 8- to 12-year-old children, the cross-sectional design does not allow conclusions regarding the direction of effects. Although theoretical considerations suggest that executive functions are fundamental skills for enacted SRL ([Dörrenbächer-Ulrich & Bregulla, 2024](#)) and pioneering empirical studies offer some evidence for such a causal link in children of

different ages (Davis et al., 2021; Rutherford et al., 2018), more longitudinal studies on the relation of executive functions, conditional strategy knowledge, and enacted SRL are needed. These would offer insight into the development of the constructs and the direction of their relations in childhood and beyond. Such studies can also inform endeavors to design interventions aimed at fostering children's strategic learning behavior. Both executive functions and SRL are promising targets for training in children of different ages (Dignath & Büttner, 2008; Titz & Karbach, 2014). Thus, an increased understanding of their relations may form the empirical basis for innovative attempts combining strategy instruction (Dignath & Büttner, 2008) with cognitive training elements (Titz & Karbach, 2014) aimed at supporting children's learning.

CRedit authorship contribution statement

Sophia Elisabeth Grobe: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Tanja Könen:** Writing – review & editing, Methodology. **Christina Rocker:** Writing – review & editing, Project administration, Data curation. **Lena Maffert-Grüneisen:** Writing – review & editing, Project administration, Data curation. **Laura Dörrenbächer-Ulrich:** Writing – review & editing, Funding acquisition. **Franziska Perels:** Writing – review & editing, Resources, Funding acquisition. **Julia Karbach:** Writing – review & editing, Resources, Funding acquisition.

Ethics statement

All procedures were in accordance with the Ethical Principles of Psychologists and Code of Conduct of the American Psychological Association (APA). This study was approved by the institutional ethics committee (LEK-436) at RPTU University Kaiserslautern-Landau.

Funding

This work was supported by the German Research Foundation [Grant numbers KA 3216/6–1, PE 1176/21–1], Bonn, Germany.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.cogdev.2025.101644](https://doi.org/10.1016/j.cogdev.2025.101644).

Data availability

Research data will be made available upon request.

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